

**Monetary Policy Forecasts and Shocks Spillovers
from the U.S to Canada and Mexico: replication and
extension of a structural VAR model.**

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1 Introduction

We replicate the main figures of Stock and Watson (2001) in section 1. In section 2, we extend the model by: i) computing supplemental tests to the original work; ii) analyzing differences within given time samples; iii) delivering new insights on spillover shocks on Canada and Mexico.

2 Replication

2.1 Model

As in the original paper, we consider a sample from 1960 to 2000 including U.S. inflation, denoted p , unemployment, denoted u , and the federal funds rate, denoted r , along with a reduced form VAR(4) (see Appendix Section A.1).

Structural shocks are identified using a recursive Cholesky decomposition with ordering $[p, u, r]$. This ordering is consistent with standard sticky-price and Taylor-rule interpretations commonly used in the SVAR literature. Further considerations of this ordering are discussed in Appendix.

One caveat is that the model is close to non-stationary boundary, which indicates that the variables of this VAR are highly persistent over time.

2.2 Preliminaries

Regarding stationarity, standard unit-root diagnostics indicate substantial persistence in the macroeconomic variables, with mixed evidence across PP, DF-GLS, and KPSS tests (Appendix A.1.2). Following the original Stock–Watson framework, variables are therefore retained in levels.

Lag-order selection criteria provide mixed recommendations across specifications (Table 1). Residual Portmanteau diagnostics indicate that low-order VARs suffer from residual autocorrelation, while intermediate lag structures substantially improve specification adequacy (Table 2). We therefore retain the VAR(4) specification used in the original article.

Table 1. Lag order specification test

Test/Lag	1	2	3	4	5	6	7	8
AIC(n)	-2.628	-3.096	-3.205	-3.188	-3.189	-3.289	-3.252	-3.278
HQ(n)	-2.532	-2.929	-2.967	-2.879	-2.808	-2.836	-2.728	-2.682
SC(n)	-2.393	-2.685	-2.619	-2.426	-2.251	-2.174	-1.962	-1.811
FPE(n)	0.072	0.045	0.041	0.041	0.041	0.037	0.039	0.038

Table 2. Series of Portmanteau tests across lag orders

Lag order	1	2	3	4	5	6	7	8
Portmanteau p-value	0	0.00023	0.00186	0.00322	0.00466	0.00050	0.00037	0.00141

2.3 Granger-Causality

A concern of this macro-model is to verify whether lagged regressors effectively contain information about the outcome vector. P-values of each test are displayed in [Table 3](#): we observe that own lagged values are highly correlated with present-values, which is trivial; past inflation do not predict significantly unemployment, which is surprising in regard of the Phillips curve, but do predict significantly the interest rate which is consistent with the Taylor rule specification; past interest rate values do not predict well inflation, which is again surprising, yet there is strong evidence of monetary impact on unemployment.

Table 3. Granger-Causality Tests (Panel A, Table 1)

	p	u	r
p	0.00	0.37	0.00
u	0.12	0.00	0.00
r	0.24	0.00	0.00

2.4 Forecast error variance decomposition

In the forecast results of each regression, we can decompose the variance contribution of each regressor, i.e how much of forecast uncertainty is due to this variable at a given horizon. The horizon schedule is set to 1 quarter, 1 year, 2 years, 3 years after the shock.

Results are displayed in [Table 4](#). Short-run inflation and unemployment are self-driven, consistent with the rigidity of these variables. Short-run interest rate is mainly influenced by past policies, with minor adjustments related to inflation and unemployment. Concerning medium-effects, the two former variables remain highly persistent, while interest rate is biased towards unemployment updates. In the long-run, unemployment is equally influenced by inflation and interest rates, and the monetary policy channel argued in the previous part.

Table 4. Variance decomposition (Panel B, Table 1)

Variance Decomposition of p			Variance Decomposition of u			Variance Decomposition of r		
p	u	r	p	u	r	p	u	r
1.00	0.00	0.00	0.00	1.00	0.00	0.03	0.18	0.79
0.90	0.08	0.02	0.00	0.97	0.02	0.12	0.48	0.41
0.86	0.12	0.02	0.08	0.79	0.13	0.15	0.57	0.29
0.86	0.12	0.02	0.21	0.61	0.19	0.20	0.55	0.26

We then assess forecast stability. While we expect that precision shrinks with horizons, it is relevant to determine the speed at which each variable uncertainty increases to. Results are displayed in [Table 5](#). Inflation and interest rate forecasts are extremely volatile relative to unemployment. For instance, this dynamic is consistent with previous tests, that shows persistent shock accumulation of inflation. Moreover, as interest rates are set ‘freely’ by the FED, this volatility is expected a priori.

Table 5. Forecast standard error decomposition (Panel B, Table 1)

p	u	r	Forecast Horizon
1	0.23	0.88	1
1.42	0.64	1.89	4
1.82	0.8	2.49	8
2.04	0.95	2.67	12

2.5 Impulse Response Functions

2.5.1 Recursive VAR

In this part, we address the main question of interest, i.e, how does each variable react to the other. Impulse response functions allows us to get a graphical interpretation of marginal dynamics across quarters. They are computed in [Figure 1](#).

Own shocks generate immediate positive responses: it can be seen for inflation, which have the highest reaction response, and interest rate; unemployment in more moderate effect. This confirms strong short-run persistence.

As for the Phillips curve relation, results are counter-intuitive. Hence, VAR regressions do not separate demand and supply shocks, which are often counter-cyclical. For example, a shock in inflation due to an oil crisis can lead firms to reduce employment to cover costs.

Interest rates are, as expected, affected directly to inflation and unemployment shocks. In the medium and long-run, variables seem to converge again to steady-state.

2.5.2 Forward-looking and backward-looking monetary rules

We compare two alternative specifications of monetary-policy identification. In the backward-looking rule, the policy rate reacts to contemporaneous observed macroeconomic conditions. In the forward-looking rule, monetary policy instead reacts to expectations of future macroeconomic conditions generated by the VAR itself.

We then replace the value of interest rate by those expectations. Impulse response functions are displayed in [Figure 2](#). Interest-rate responses remain unchanged because the VAR is backward-looking. Inflation and unemployment responses are smoother, which is consistent with the rational behavior of agents. Formal explanation is detailed in Section A.2.1.

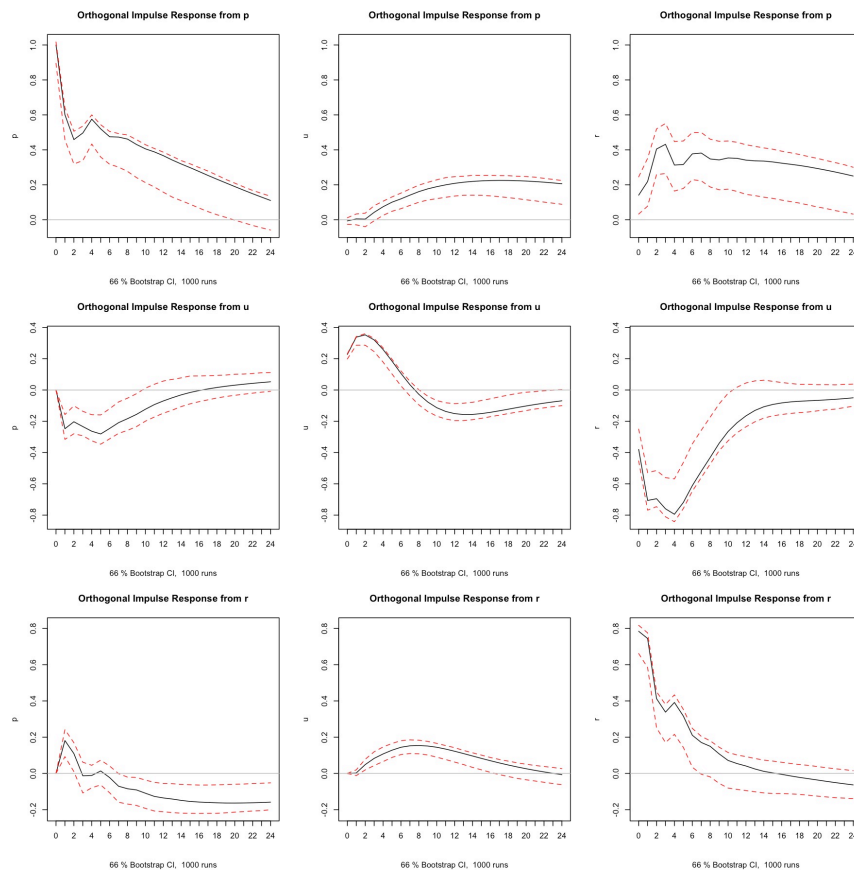


Figure 1. Impulse response functions (Figure 1)

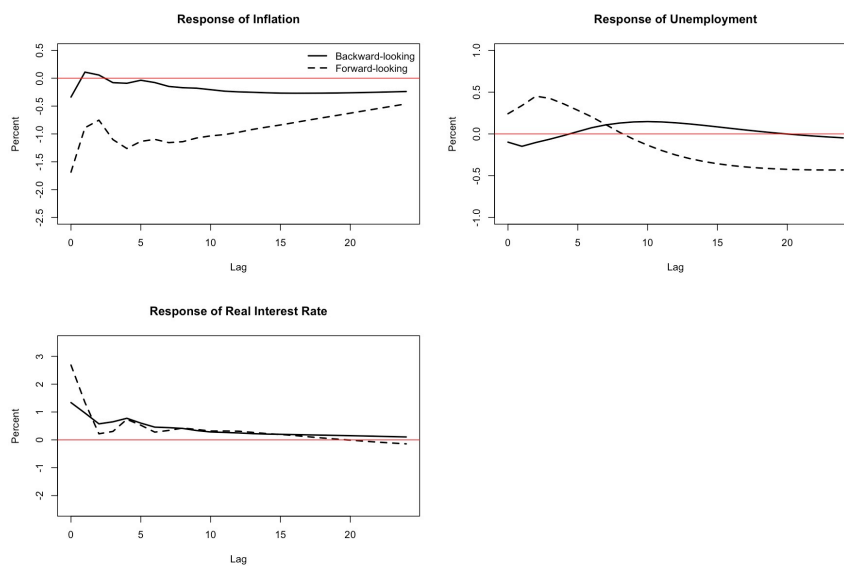


Figure 2. Impulse response functions for different monetary rules (Figure 2)

3 Extensions

3.1 Structural break of the 2008 financial crisis

“Did monetary-policy transmission change after the financial crisis, holding roughly comparable modern monetary-policy regimes?”. Firstly on sample construction, quarterly GDPDEF prevented meaningful post-crisis subsample analysis, hence was replaced by monthly CPI. We define two periods : pre-crisis from 1984 Q1 to 2007 Q4 and post-crisis 2010 Q1 to 2019 Q4 to exclude the 2008-2009 break and the COVID pandemic.

3.1.1 Stationarity and lag order selection

Stationarity diagnostics indicate substantial persistence across variables and subsamples, particularly for unemployment and interest rates. Relative to the pre-crisis sample, post-crisis results also suggest moderately stronger persistence across all variables, especially for inflation and interest rates. Detailed stationarity diagnostics are reported in Appendix A.

Lag-selection criteria consistently favor a VAR(2) specification before 2008 and a VAR(4) specification afterward. Residual Portmanteau diagnostics confirm that higher-order post-crisis dynamics are necessary to adequately remove serial correlation ([Table 12](#)). We therefore retain a VAR(2) pre-crisis and a VAR(4) post-crisis specification.

Table 6. Lag-order selection criteria across pre- and post-crisis subsamples

Sam- ple	Crite- rion	1	2	3	4	5	6	7	8	9	10
Pre	AIC(n)	-5.329	-5.706	-5.672	-5.655	-5.626	-5.628	-5.614	-5.576	-5.562	-5.568
Pre	HQ(n)	-5.266	-5.595	-5.514	-5.450	-5.373	-5.328	-5.267	-5.182	-5.120	-5.079
Pre	SC(n)	-5.172	-5.431	-5.279	-5.144	-4.996	-4.880	-4.749	-4.593	-4.460	-4.348
Pre	FPE(n)	0.005	0.003	0.003	0.003	0.004	0.004	0.004	0.004	0.004	0.004
Post	AIC(n)	-7.856	-8.037	-7.984	-8.193	-8.110	-8.003	-7.995	-7.911	-7.879	-7.797
Post	HQ(n)	-7.736	-7.826	-7.682	-7.800	-7.627	-7.429	-7.330	-7.155	-7.033	-6.860
Post	SC(n)	-7.558	-7.516	-7.239	-7.224	-6.918	-6.587	-6.356	-6.048	-5.793	-5.487
Post	FPE(n)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

3.1.2 Monetary-policy transmission across subsamples

We compare recursively identified monetary-policy shocks across pre and post-crisis subsamples. Before 2008, contractionary interest-rate shocks generate dynamics broadly consistent with the original Stock-Watson framework: inflation gradually decreases and interest rates smoothly revert toward equilibrium. Unemployment responses, however, display a classical unemployment puzzle, with contractionary shocks associated with declining unemployment.

Post-crisis responses appear weaker and more persistent, particularly for inflation, whose adjustment becomes slower, while unemployment remains elevated for longer periods. These results suggest weaker conventional monetary-policy transmission after 2008, although interpretation remains cautious because the Zero Lower Bound regime mechanically reduced the variability of conventional interest-rate shocks.

Additional robustness checks based on the complete impulse-response system are reported in Appendix A.3.1.2.

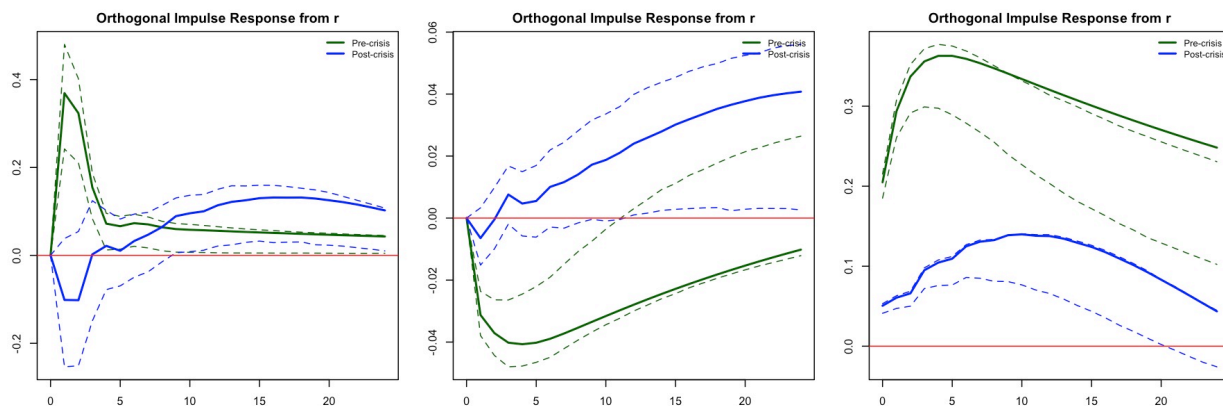
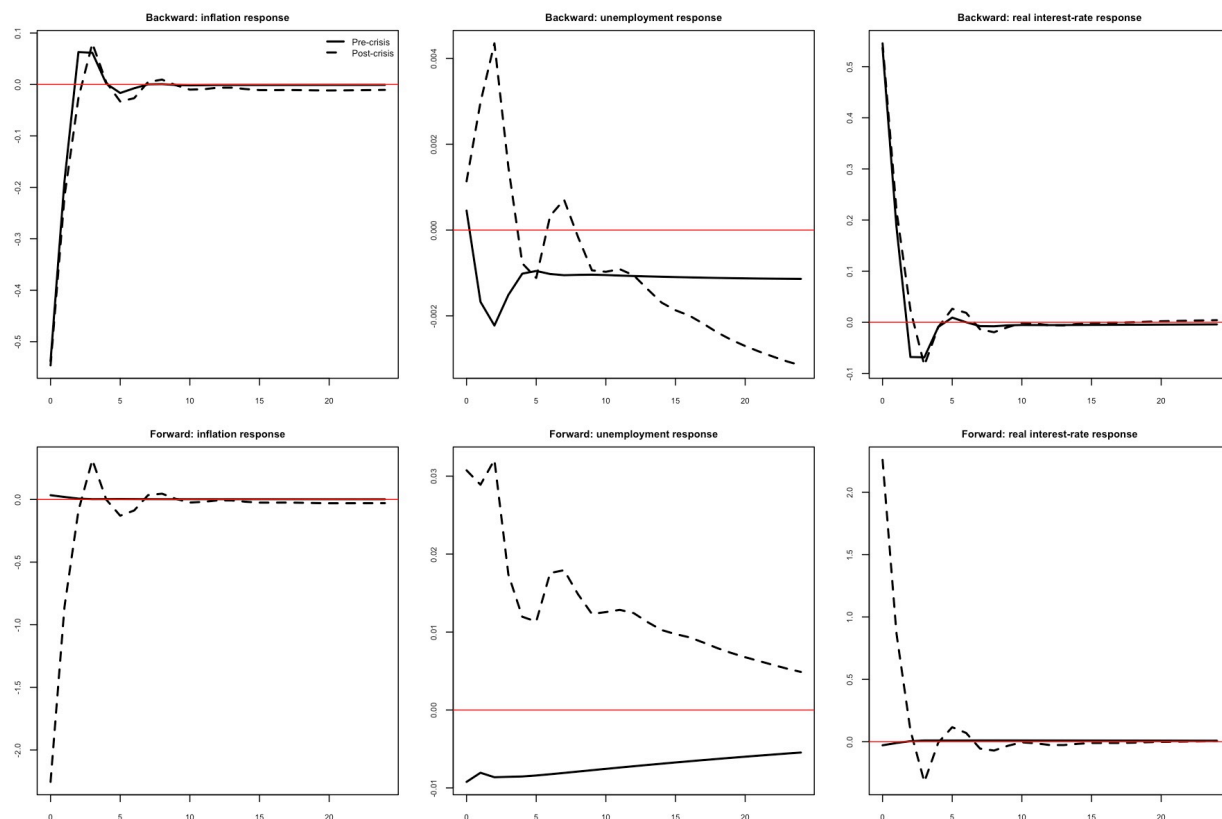


Figure 3. Comparison of responses of inflation, unemployment and the interest rate to a one-standard-deviation monetary-policy shock in r

We also reproduce the backward and forward-looking Taylor-rule identifications. The backward-looking specification delivers responses broadly consistent with the recursive benchmark for inflation and real interest rates, but unemployment responses remain fragile across subsamples and occasionally exhibit sign reversals. The forward-looking specification is substantially less stable. Because the fixed-point correction term approaches singularity in the pre-crisis sample,

the estimated policy coefficients become mechanically amplified, generating implausibly large responses, particularly for unemployment. Consequently, forward-looking Taylor-rule identification becomes highly sensitive to persistence, inflation volatility, and small-sample estimation in monthly data.



We therefore retain the recursive SVAR as the benchmark specification. Despite their greater instability, the Taylor-rule specifications broadly support the evidence of more persistent post-crisis monetary-policy transmission.

3.2 Spillover effects

Considering the influence of the U.S. on the global economy, we study the spillover effects of U.S. monetary policy on Canada and Mexico. These two countries provide an informative comparison because they follow different macroeconomic trajectories.

3.2.1 Model specification

We use a quarterly sample from Mexico and Canada ranging from 1960 to 2019 for Canada and from 1990 to 2019 for Mexico. This asymmetry is due to a lack of reliable data sources, which lowers the interpretation credibility of the latter. However, further tests in the appendix shows that results are robust to this specification, making us move forward.

When computing stationarity, lag-order specification and other inference tests, we argue that spillover effects only are unsurprisingly relevant for low-orders. Hence, we decide to use a VAR(2) specification. As in the previous part, we also compared between a restricted and a full model, that suggested a higher degree of precision when using the restricted specification.

3.2.2 Spillovers dynamics

Impulse responses in [Figure 5](#) and forecast-error variance decompositions in [Figure 6](#) and [Figure 7](#) reveal heterogeneous spillover dynamics across countries. Canada exhibits moderate financial spillovers: domestic interest rates react significantly to U.S. monetary-policy shocks, while effects on unemployment remain limited. Mexico displays substantially stronger dependence on U.S. macroeconomic conditions, particularly through inflation and interest-rate dynamics, consistent with the greater vulnerability of emerging economies to external financial conditions. By contrast, both models show weak spillovers through unemployment channel, which is expected because we may argue that job market frictions do take time.

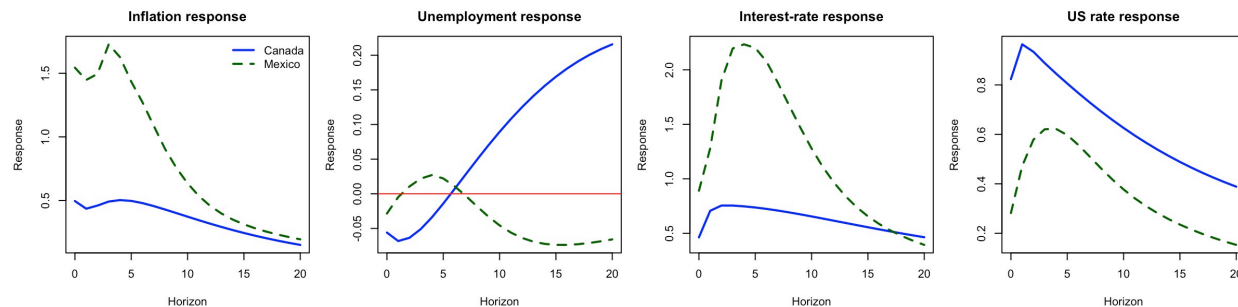


Figure 5. Orthogonal IRF on CAN and MEX (Restricted)

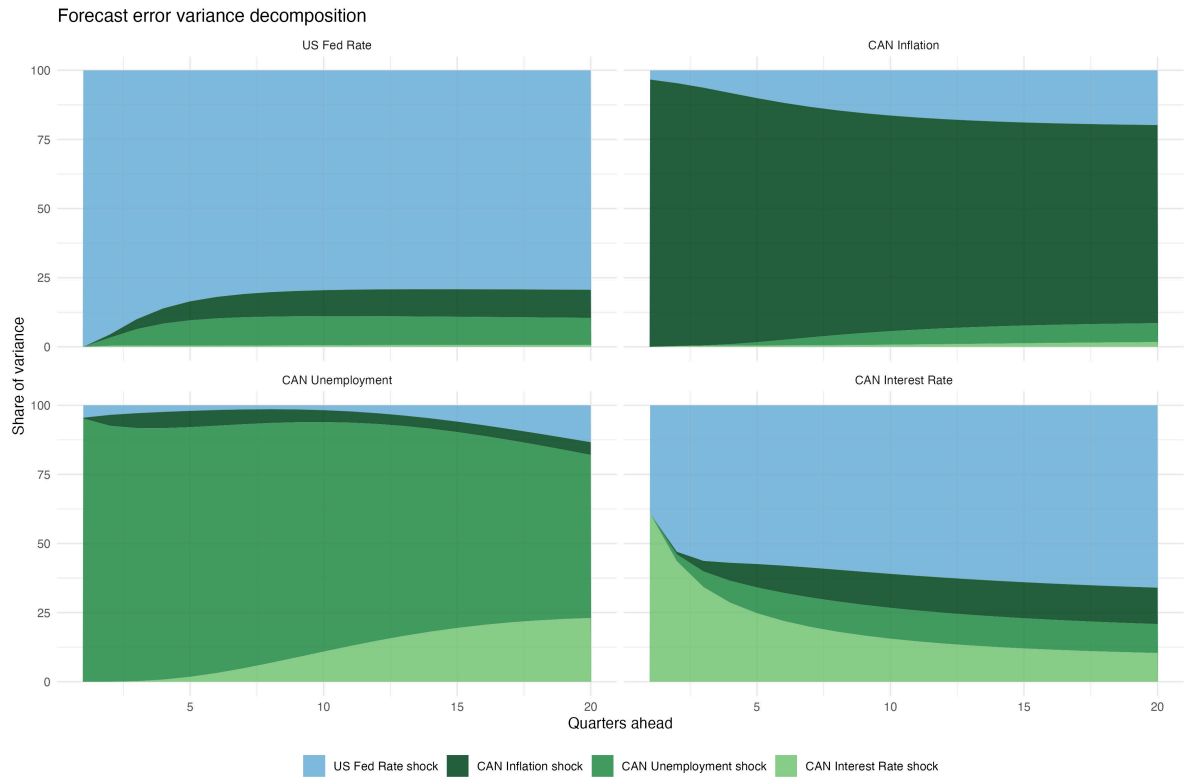


Figure 6. Forecast Variance Decomposition (FEVD) plot of Canada (Quarterly, Restricted)

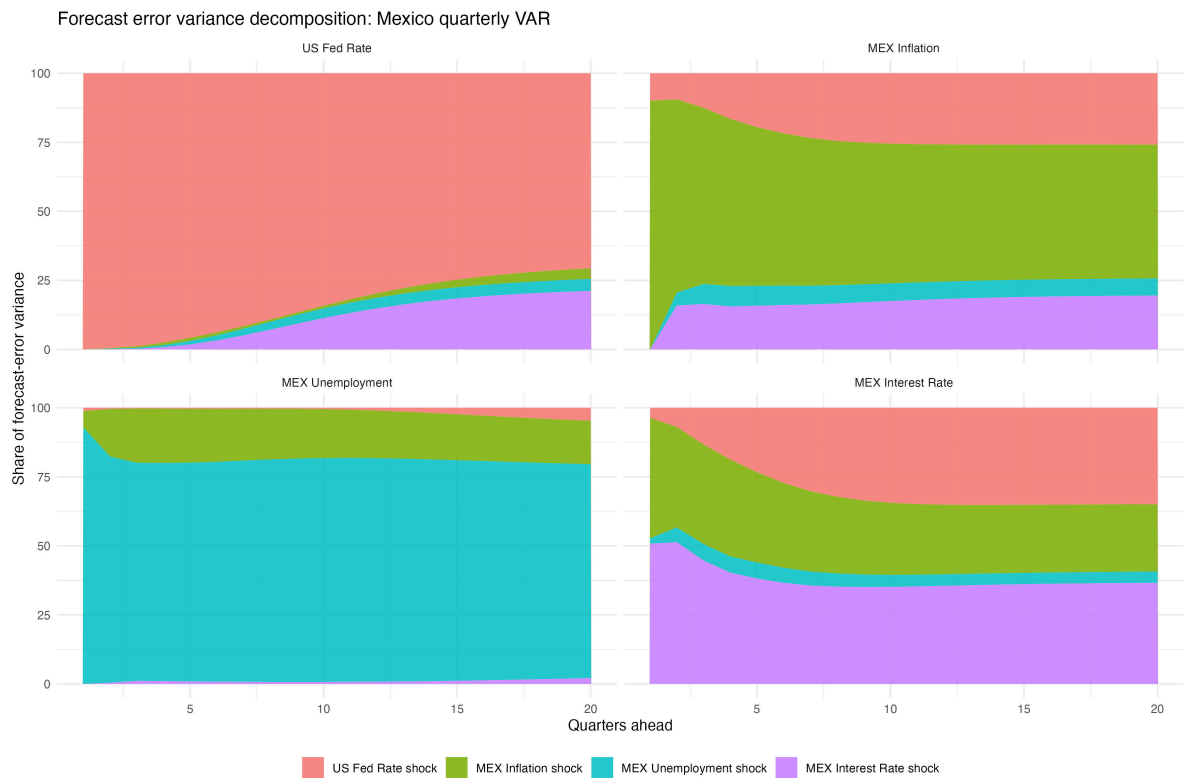


Figure 7. Forecast Variance Decomposition (FEVD) plot of Mexico (Quarterly, Restricted)

4 Conclusion

This paper replicates the structural VAR framework of Stock and Watson and confirms the persistent and gradual effects of monetary-policy shocks. The post-crisis extension suggests that monetary-policy transmission became more persistent after 2008, although identification is weakened by the ZLB regime and by the sensitivity of Taylor-rule specifications to persistence and sample choice. The international extension shows moderate U.S. monetary-policy spillovers for Canada but substantially stronger effects for Mexico, consistent with the greater sensitivity of emerging economies to U.S. monetary conditions.

Overall, the analysis emphasizes the importance of persistence, identification assumptions, and specification choices in structural VAR analysis.

A Appendix

A.1 Preliminaries

A.1.1 Cholesky ordering

The ordering of variables is crucial as we apply Cholesky ordering to estimate structural shocks within variables. Hence, the first variable is said to be contemporaneously exogenous, whilst the others are recursively impacted by the variables put before. In this context, we argue that inflation cannot change in the short-run (e.g. prices are sticky or menu-costs), unemployment are set to be moving (inversely) to inflation (i.e. the Phillips Curve), and interest rates are decided upon these two dynamics, which is plausible assuming the FED sets its rate according to a special rule of inflation target. Alternative orderings reported thereafter produce very similar IRFs, suggesting that the main results are not driven by this contemporaneous ordering assumption.

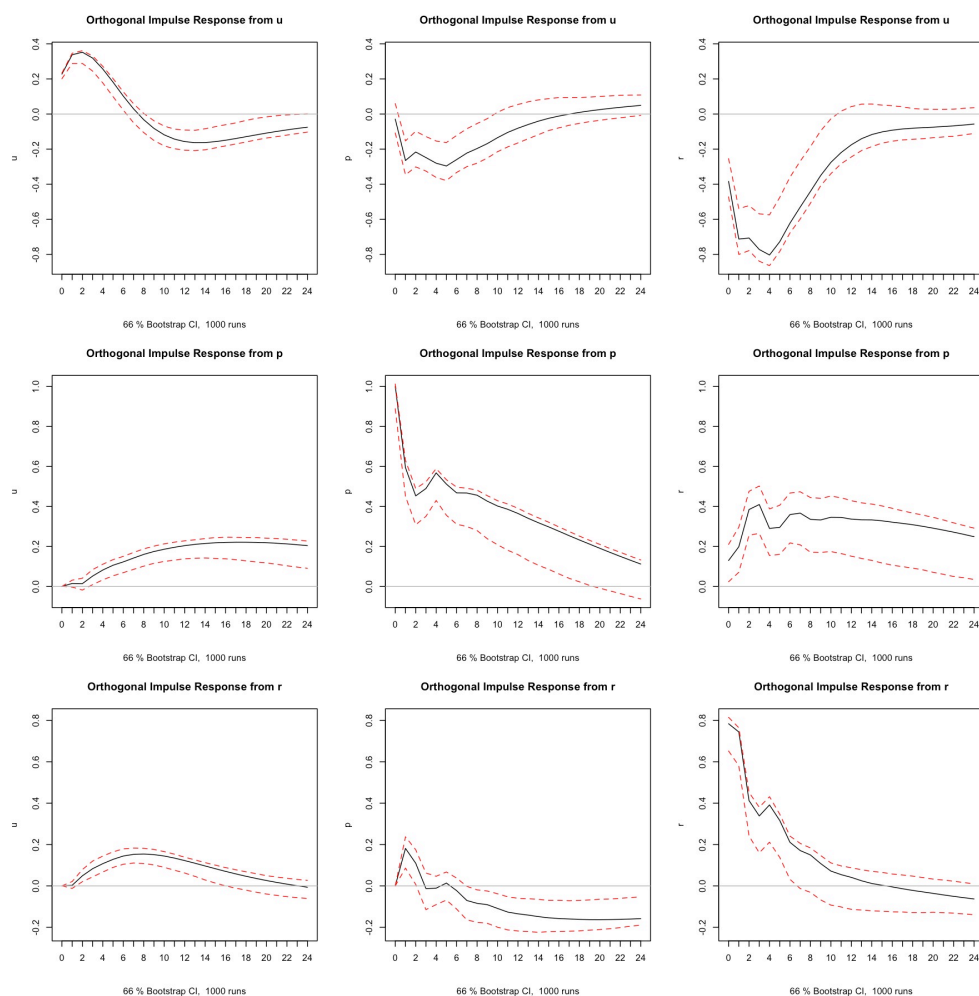


Figure 8. IRF functions for the (u,p,r) ordering

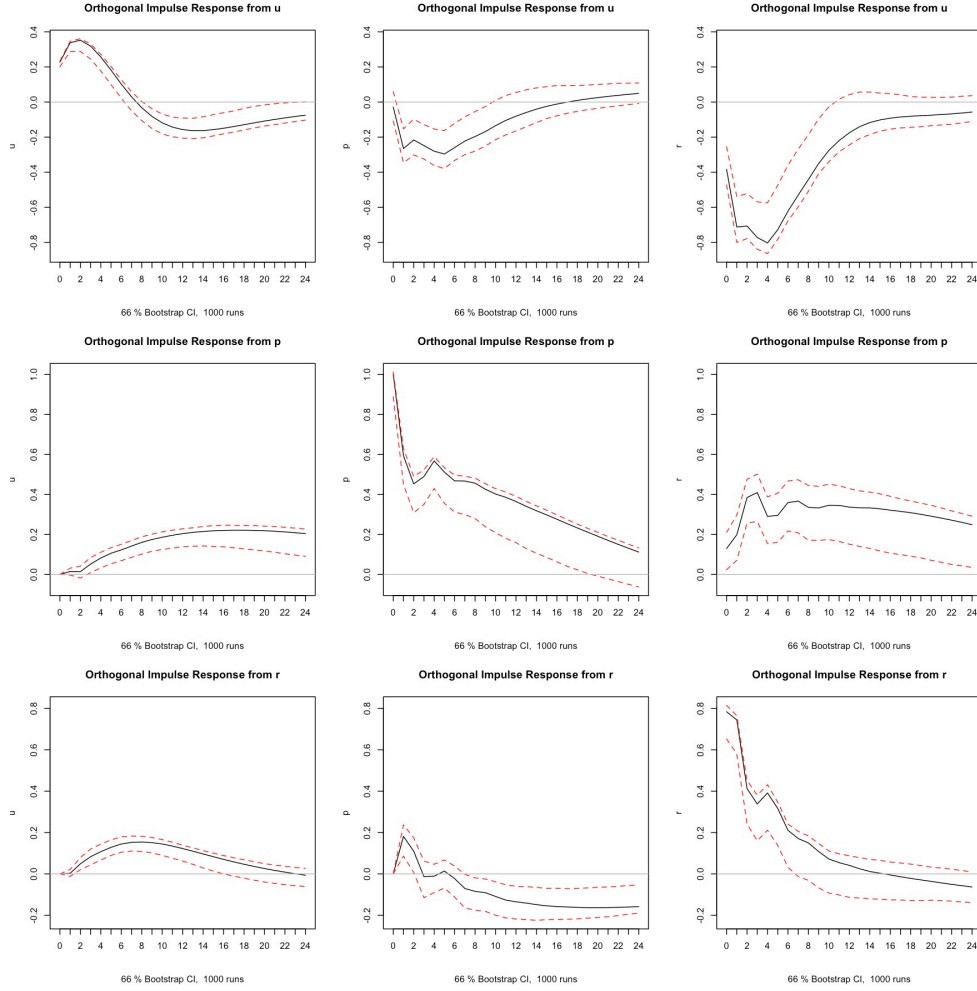


Figure 9. IRF functions for the (r,p,u) ordering

A.1.2 Stationarity

Non-stationarity tests reported in Table 9 provide mixed evidence. Phillips-Perron tests fail to reject unit roots, while KPSS tests reject stationarity for all variables. DF-GLS statistics provide partial evidence against unit roots for unemployment and interest rates. Overall, the variables appear highly persistent. The characteristic roots reported in Table 10 all lie inside the unit circle, implying that the benchmark VAR(4) remains dynamically stable, although the largest roots remain close to one.

Table 7. PP/DFGLS/KPSS non stationarity test results

Variable	PP stat	PP p-value	DFGLS stat	DFGLS cv(5%)	KPSS stat	KPSS p-value
p	-12.766	0.383	-1.678	-1.94	0.694	0.014
u	-8.488	0.628	-1.971	-1.94	0.508	0.04
r	-11.342	0.465	-2.087	-1.94	0.585	0.024

The Phillips–Perron test implemented through `tseries::pp.test()` reports the Dickey–Fuller $Z(\alpha)$ statistic rather than the standard ADF tau statistic. Consequently, strongly negative statistics do not mechanically imply small p-values.

Table 8. Unit-root results

Root	1	2	3	4	5	6	7	8	9	10	11	12
Mod- ulus root	0.971	0.971	0.803	0.803	0.685	0.685	0.571	0.571	0.446	0.446	0.190	0.190

A.1.3 Lag order specification

Because the original paper does not justify lag selection, we compare specifications using information criteria and residual diagnostics. Results reported in Table 11 provide mixed recommendations: Schwarz/BIC favors parsimonious models such as VAR(3), whereas AIC, HQ, and FPE favor richer lag structures around VAR(6).

To further discriminate between specifications, we compute asymptotic Portmanteau serial-correlation tests across lag orders. Results reported in Table 10 strongly reject residual independence for low-order VARs, indicating that small lag structures fail to capture the persistence of the system. Although residual autocorrelation remains statistically significant across specifications, higher lag orders moderately improve the diagnostics.

We therefore focus on VAR(3), VAR(4), and VAR(5), all of which satisfy stability conditions (Table 11). Figure 10 shows robust impulse responses across lag specifications.

We ultimately retain the VAR(4) specification used in the original article. Relative to VAR(3), it provides greater dynamic flexibility, while VAR(5) yields nearly identical impulse responses at the cost of additional parameterization. VAR(4) therefore represents a reasonable compromise between parsimony, flexibility, and stability.

Table 9. Lag order specification test

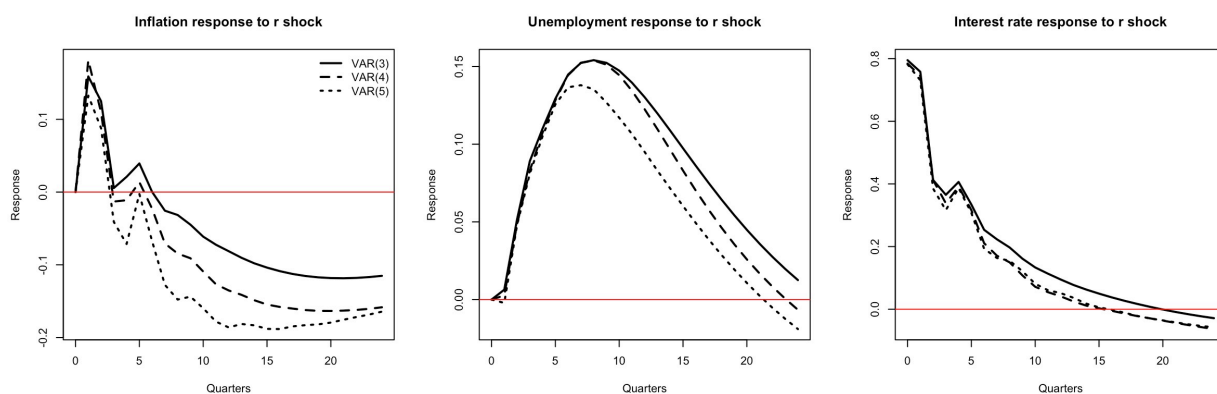
Test/Lag	1	2	3	4	5	6	7	8
AIC(n)	-2.628	-3.096	-3.205	-3.188	-3.189	-3.289	-3.252	-3.278
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SC(n)	-2.393	-2.685	-2.619	-2.426	-2.251	-2.174	-1.962	-1.811
FPE(n)	0.072	0.045	0.041	0.041	0.041	0.037	0.039	0.038

Table 10. Series of Portmanteau test in function of lag orders

Lag order	Portmanteau p-value
1	0
2	0.00023
3	0.00186
4	0.00322
5	0.00466
6	0.00050
7	0.00037
8	0.00141

Table 11. Maximum stability roots across lag specifications

Model	Maximum root modulus
VAR(3)	0.958
VAR(4)	0.971
VAR(5)	0.964

**Figure 10.** Comparison of different impulse response functions

A.2 Replication

A.2.1 Mathematical appendix: forward-looking Taylor rule¹

Let a monetary rule defined as:

$$r_t = f_p^0 \cdot \mathbb{E}_t[\overline{p_{t+k}}] + f_u^0 \cdot \mathbb{E}_t[\overline{u_{t+k}}]$$

¹This part was induced from the original replication package of the paper.

with f_p^0 and f_u^0 target weights of inflation and unemployment respectively (here for example, authors set $f_p^0 = 1.5$ and $f_u^0 = -1.25$). $\overline{p_{t+k}} (\overline{u_{t+k}})$ defines the rolling average of inflation (unemployment) at date t for k quarters ahead. We form again a structural VAR such as:

$$Z_t = \Phi Z_{t-1} + \varepsilon_t$$

with $Z_t' = [p_t, u_t, r_t, p_{t-1}, u_{t-1}, r_{t-1}, \dots, r_{t-3}] \in \mathbb{R}^{1 \times 12}$ and providing that $|\max(\Phi)| < 1$. We then have:

$$\mathbb{E}_t[Z_{t+k}] = \Phi^k Z_t$$

assuming that $\mathbb{E}_t[\varepsilon_{t+i}] = 0 \ \forall i \in \mathbb{Z}$ such that, for a given variable X at time t ,

$$\mathbb{E}_t[\overline{X_{t+k}}] = \frac{1}{k} \sum_{i=1}^k \Phi^i X_t$$

which gives us:

$$r_t = f_p^0 \cdot e'_p \left(\frac{1}{k} \sum_{i=1}^k \Phi^i Z_t \right) + f_u^0 \cdot e'_u \left(\frac{1}{k} \sum_{i=1}^k \Phi^i Z_t \right) = (f_p^0 \cdot e'_p + f_u^0 \cdot e'_u) \underbrace{\frac{1}{k} \sum_{i=1}^k \Phi^i Z_t}_{:= F_k}$$

with $e'_p = (1, 0, \dots, 0)$, $e'_u = (0, 1, 0, \dots, 0)$. We can also set

$$F_k = \begin{pmatrix} \frac{1}{k} \sum_i^k \Phi_{[p,p]}^i & \frac{1}{k} \sum_i^k \Phi_{[p,u]}^i & \frac{1}{k} \sum_i^k \Phi_{[p,r]}^i & \frac{1}{k} \sum_i^k \Phi_{[p,p-1]}^i & \dots \\ \frac{1}{k} \sum_i^k \Phi_{[u,p]}^i & \frac{1}{k} \sum_i^k \Phi_{[u,u]}^i & \frac{1}{k} \sum_i^k \Phi_{[u,r]}^i & \frac{1}{k} \sum_i^k \Phi_{[u,p-1]}^i & \dots \\ \frac{1}{k} \sum_i^k \Phi_{[r,p]}^i & \frac{1}{k} \sum_i^k \Phi_{[r,u]}^i & \frac{1}{k} \sum_i^k \Phi_{[r,r]}^i & \frac{1}{k} \sum_i^k \Phi_{[r,p-1]}^i & \dots \\ \frac{1}{k} \sum_i^k \Phi_{[p-1,p]}^i & \frac{1}{k} \sum_i^k \Phi_{[p-1,u]}^i & \frac{1}{k} \sum_i^k \Phi_{[p-1,r]}^i & \frac{1}{k} \sum_i^k \Phi_{[p-1,p-1]}^i & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{pmatrix} = \begin{pmatrix} a_{pp} & a_{up} & a_{rp} & \dots \\ a_{pu} & a_{uu} & a_{ru} & \dots \\ a_{pr} & a_{ur} & a_{rr} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

We can decompose for contemporaneous and lagged responses:

$$r_t = \underbrace{(f_p^0 \cdot e'_p + f_u^0 \cdot e'_u)_{1:3} F_k Z_t}_{\text{contemporaneous}} + \underbrace{(f_p^0 \cdot e'_p + f_u^0 \cdot e'_u)_{4:12} F_k Z_t}_{\text{lags}}$$

Given $F_k = \begin{pmatrix} a_{pp} & a_{up} & a_{rp} & \dots \\ a_{pu} & a_{uu} & a_{ru} & \dots \\ a_{pr} & a_{ur} & a_{rr} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$ and a_{xy} effect of current y on the expected prediction coefficient

of x , we have that $f_p^0 e'_p F_k = f_p^0 (a_{pp}, a_{up}, a_{rp}, \dots)$ and $f_u^0 e'_u F_k = f_u^0 (a_{pu}, a_{uu}, a_{ru}, \dots)$ that gives the vector $(f_p^0 \cdot e'_p + f_u^0 \cdot e'_u) F_k = (f_p^0 a_{pp} + f_u^0 a_{pu}, f_p^0 a_{up} + f_u^0 a_{uu}, \dots)$ such that, recalling that $Z_t' = [p_t, u_t, r_t, p_{t-1}, u_{t-1}, r_{t-1}, \dots, r_{t-3}]$ and that we can define $Z_t'^{(3)} = [p_t, u_t, r_t]$ the first three elements of the variables vector and with a slight abuse of notation $Z_t'^{(-3)}$ the remaining elements,

$$r_t = (f_p^0 \cdot e'_p + f_u^0 \cdot e'_u) F_{k_{[1:3]}} Z_t'^{(3)} + (f_p^0 \cdot e'_p + f_u^0 \cdot e'_u) F_{k_{[4:12]}} Z_t'^{(-3)}$$

$$\Leftrightarrow r_t = (f_p^0 a_{pp} + f_u^0 a_{up})p_t + (f_p^0 a_{pu} + f_u^0 a_{uu})u_t + (f_p^0 a_{pr} + f_u^0 a_{ur})r_t + \text{lags}$$

$$\Leftrightarrow r_t = \underbrace{\frac{f_p^0}{1 - (f_p^0 a_{pr} + f_u^0 a_{ur})}}_{f_p^{\text{fwd}}} p_t + \underbrace{\frac{f_u^0}{1 - (f_p^0 a_{pr} + f_u^0 a_{ur})}}_{f_u^{\text{fwd}}} u_t + \text{lags}$$

We deduce finally a recursive VAR that is identifiable such that:

$$ra_t := r_t - \underbrace{\frac{f_p^0}{1 - (f_p^0 a_{pr} + f_u^0 a_{ur})} p_t - \frac{f_u^0}{1 - (f_p^0 a_{pr} + f_u^0 a_{ur})} u_t}_{Z_t} = \underbrace{\Gamma \text{lags}}_{Z_{t-1}} + \varepsilon_t$$

Empirically, we have $f_u^{\text{bwd}} = \frac{-1.25}{4}$, $f_p^{\text{bwd}} = \frac{1.5}{4}$ (recall that we take rolling means across 4 quarters), and $f_u^{\text{fwd}} = -3.359$, $f_p^{\text{fwd}} = 0.757$ (according to the replication package). We notice directly that $|f_u^{\text{bwd}}| < |f_u^{\text{fwd}}|$ and that $|f_p^{\text{bwd}}| < |f_p^{\text{fwd}}|$. Hence, one way to explain that the FED puts more weights on this forward-looking framework is that they do expect that it does expect now that future values of unemployment and inflation are endogenous on interest rates values, whereas it was determined before.

A.3 Extensions

A.3.1 Subsamples pre and post crisis

A.3.1.1 Stationarity and lag order selection

Results in Figure 11 and 12 show substantial persistence across all variables and subsamples. Inflation appears comparatively more stationary than unemployment and interest rates: Phillips–Perron and DF–GLS tests partially reject unit roots, while KPSS results improve after 2008, consistent with a more stable post-crisis inflation environment. By contrast, unemployment and federal funds rates remain highly persistent, especially in the post-crisis sample, where unit-root tests generally fail to reject non-stationarity and KPSS tests reject stationarity. Although both retained VARs satisfy stability conditions, the post-crisis VAR(4) exhibits a larger maximum root modulus (Figure 13) than the pre-crisis VAR(2), indicating stronger persistence.

variable	PP_stat	PP_pvalue	DFGLS_stat	DFGLS_cv5	KPSS_stat	KPSS_pvalue
p	-174.721	0.010	-1.946	-1.94	0.635	0.019
u	-6.175	0.764	-0.214	-1.94	2.317	0.010
r	-7.314	0.700	-0.605	-1.94	2.604	0.010

Figure 11. Stationarity results for the pre-crisis subsample

variable	PP_stat	PP_pvalue	DFGLS_stat	DFGLS_cv5	KPSS_stat	KPSS_pvalue
p	-65.544	0.010	-3.445	-1.94	0.119	0.10
u	-2.184	0.962	1.840	-1.94	2.462	0.01
r	-3.712	0.901	-1.480	-1.94	1.763	0.01

Figure 12. Stationarity results for the post-crisis subsample

sample	max_root
Pre-crisis VAR(2)	0.981
Post-crisis VAR(4)	0.993

Figure 13. Roots' comparison

Table 12. Portmanteau test p-values across lag orders

Lag	1	2	3	4	5	6	7	8	9	10
Pre-crisis	0	0.00121	0.00129	0.00267	0.00272	0.00979	0.00565	0.00169	0.00471	0.00296
Post-crisis	0.04396	0.00769	0.00658	0.6568	0.70546	0.67621	0.61795	0.30766	0.2525	0.20607

A.3.1.2 IRFs with recursive benchmark

As an additional robustness check, Figure 14 reports the complete set of orthogonal impulse responses for the pre- and post-crisis VARs. Own-variable responses remain broadly similar across subsamples, indicating that the core transmission mechanisms are preserved. At the same time, post-crisis responses generally display slower adjustment and greater persistence, particularly for inflation and interest rates. By contrast, cross-variable responses involving unemployment are more sensitive to sample choice and occasionally display economically ambiguous patterns. Overall, the full impulse-response system supports the robustness of the benchmark findings and suggests that the differences between subsamples primarily concern the strength of transmission rather than its qualitative direction.

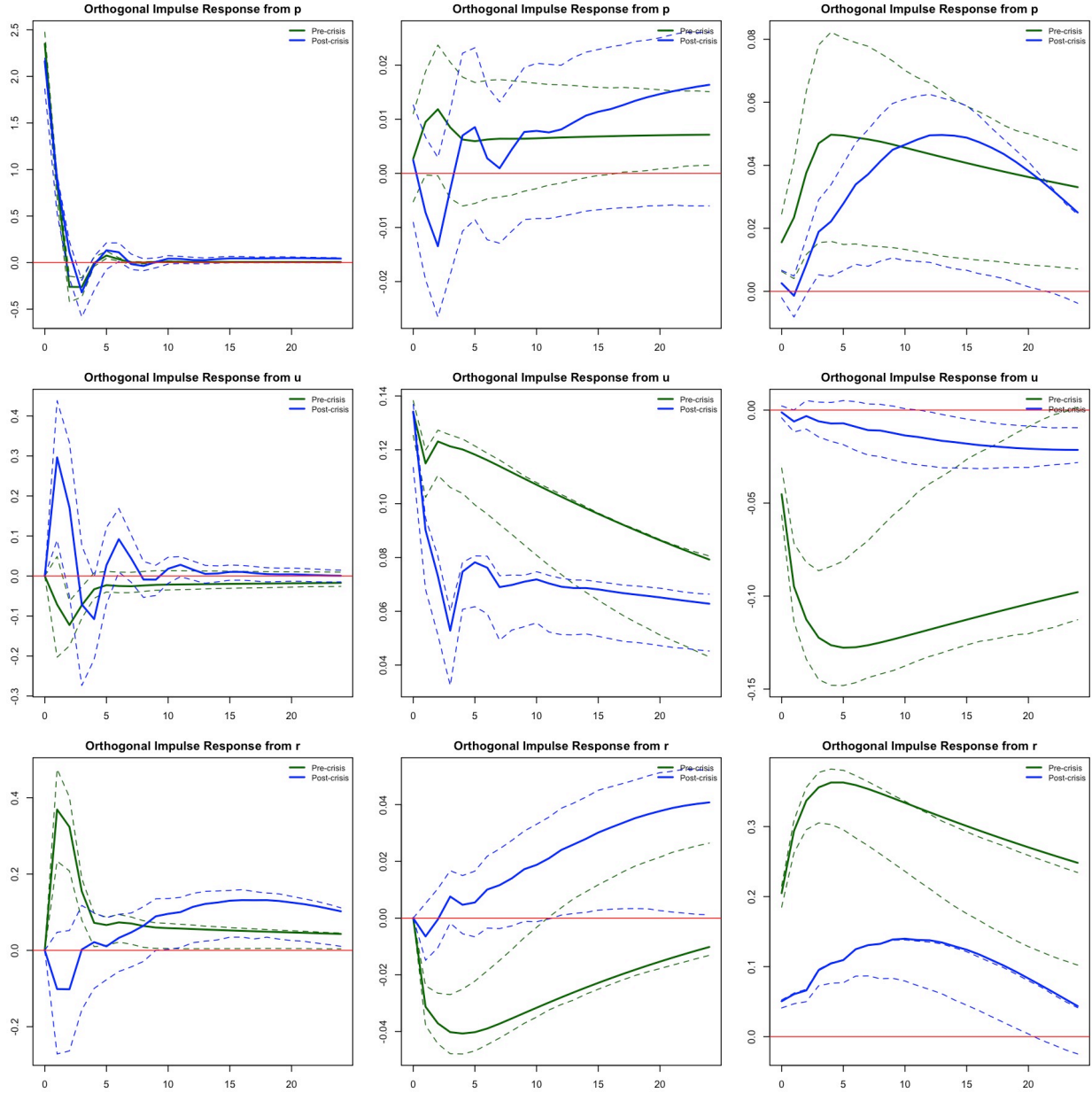


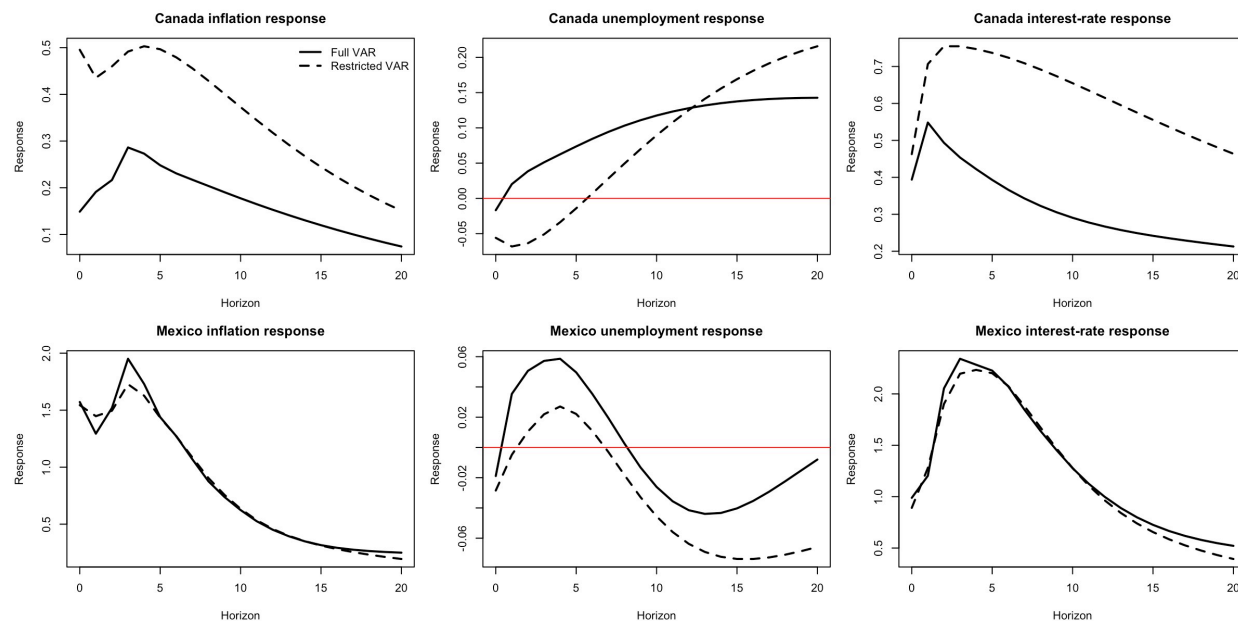
Figure 14. IRF orthogonal response functions of pre-crisis and post-crisis

A.3.2 Spillover effects analysis

We compute as for previous models Granger causality tests for all model specifications. Results in the below table suggest first evidence of spillover effects, as U.S. variables are said to be Granger correlated with domestic variables. One model though fails to reject the null.

Table 13. Granger causality test p-values table

Model	Cause	Statistic	P_value	DF
US-CAN	p,u,r	4.165	0	18
US-CAN-REST	r	3.704	0.001	1296
US-MEX-2001	p,u,r	2.261	0.002	6
US-MEX-REST	r	1.562	0.157	880

**Figure 15.** Comparison between full and limited VAR models impulse response functions.

The dynamics between the restricted model and the full model are relatively similar, which indicates that we may use both models indifferently. However, it is shown in the below tables of mean squared errors that the restrictive model is expected to be more precise over-time, which convinced thus to use it for our computations.

Table 14. Mean Squared Errors of restricted VAR (2) and full VAR (1) for Canada

Horizon	Mean Squared Errors (1)	Mean Squared Errors (2)
1	9.39	9.462
4	17.417	15.699
8	24.682	21.522
12	24.969	27.78

Table 15. Mean Squared Errors of restricted VAR (2) and full VAR (1) for Mexico

Horizon	Mean Squared Errors (1)	Mean Squared Errors (2)
1	20.561	15.485
4	60.968	34.433
8	112.121	49.841
12	142.073	57.168

As for the lag order selection, results are more convincing for Canada than for Mexico. Hence, as we could harmonize between multiple sources for the former, we have more observations. Lag order results suggest we should use a lag of order 2 for Canada but it is still dubious for Mexico. The Quarterly model suggest a VAR(2), which is consistent with previous considerations, whereas the Monthly data suggest a VAR(2) according to SC, a VAR(5) for the AIC, and VAR(2), VAR(3), VAR(4) for the HQ test. As we try to assess spillover effects of foreign monetary policy, we should not expect a high-level lag order. Hence, we choose a VAR(2) specification.

Table 16. Lag-order inference results for the CAN VAR (Quarterly, Restricted)

	1	2	3	4	5	6	7	8
AIC(n)	-1.715	-2.076	-2.072	-2.014	-2.046	-2.039	-2.021	-2.022
HQ(n)	-1.591	-1.854	-1.751	-1.595	-1.527	-1.422	-1.306	-1.208
SC(n)	-1.409	-1.526	-1.278	-0.975	-0.762	-0.511	-0.249	-0.006
FPE(n)	0.18	0.125	0.126	0.134	0.13	0.131	0.133	0.134

Table 17. Lag-order inference results for the MEX VAR (Monthly, Restricted)

	1	2	3	4	5	6	7
AIC(n)	-6.763	-7.505	-7.61	-7.679	-7.7	-7.771	-7.784
HQ(n)	-6.676	-7.348	-7.383	-7.383	-7.333	-7.335	-7.279
SC(n)	-6.544	-7.11	-7.04	-6.934	-6.78	-6.675	-6.514
FPE(n)	0.001	0.001	0	0	0	0	0

Table 18. Lag-order inference results for the MEX VAR (Quarterly, Restricted)

	1	2	3	4	5	6	7
AIC(n)	-1.618	-3.308	-3.312	-3.041	-2.825	-2.551	-2.679
HQ(n)	-1.165	-2.465	-2.081	-1.421	-0.816	-0.154	0.106
SC(n)	-0.496	-1.224	-0.267	0.966	2.144	3.379	4.212
FPE(n)	0.199	0.037	0.038	0.051	0.069	0.1	0.103